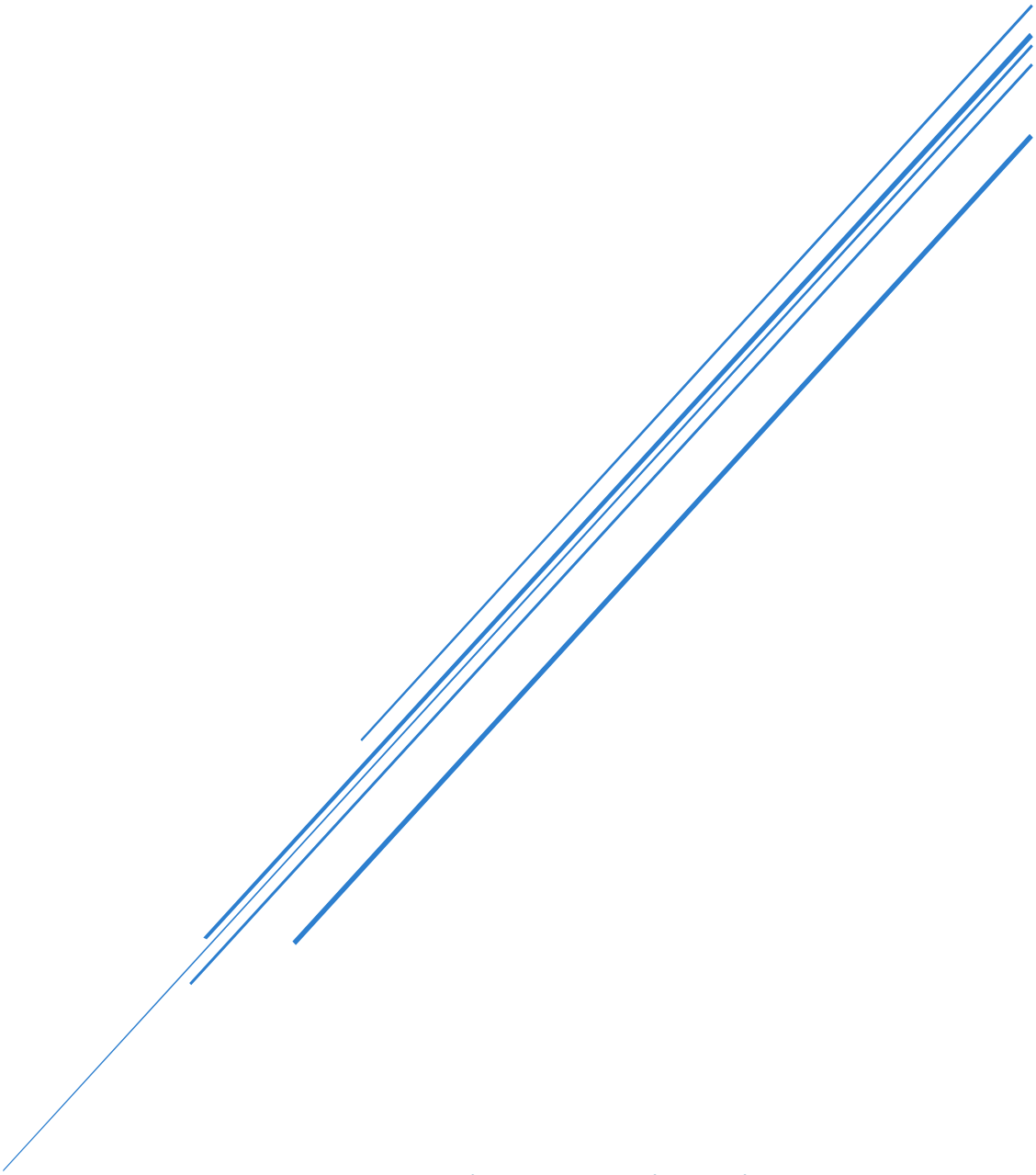
LAB 10 ANSWERS REPORT

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Date of Completion: 27/10/2026



Flinders University – Tonsley
Artificial Intelligence (COMP3742)

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Exercises

Using the traffic knowledge base defined in Source/traffic.pl, the following queries and results.

Query: [traffic].
Result: true.

Query: is_typical_traffic(morning, low).
Result: true.

Query: is_typical_traffic(afternoon, high).
Result: false.

Query: is_abnormal_traffic(evening, low).
Result: true.

Query: is_abnormal_traffic(night, moderate).
Result: true.

Discussion Question

Investigate how Prolog and logic programming can be integrated into Python machine learning software such as scikit-learn and Pytorch.

This can be achieved by using an intermediary library/module in Python such as pyswip which connects to an underlying Prolog runtime. Similar idea to what's being done in A2 for this class where scikit-learn and/or PyTorch take your features in and produce a classification or regression output that is then passed to the Prolog interface through pyswip. Then your knowledgebase can use rules and facts to produce a final output through if-then-else logic.

Questions

All the rules, facts, and if-then-else conditional statements are contained in the Source/expersystem.pl file. This expert system is a simple one for deciding to go for a run or not based on several environmental factors.

This covers lab question 1-3 (including the procedural programming style if-then-else).

Below are some demonstration input/output snippets to show use of the following: negation and other operators, if-then-else conditional logic powered by the procedural style template, nested if-then-else logic, and use of input variables.

```
37 ?- safe_weather(sunny).
true.
```

```
36 ?- safe_weather(storm).
false.
```

These two above make use of negation logic by checking if the weather is NOT unsafe weather.

```
38 ?- go_for_run(dry, 25, 9, sunny).
Go for a run, but it is a bit hot
true.

39 ?- go_for_run(dry, 25, 9, scorching).
It is too hot and the weather is not optimal, do not run
true.
```

These two make use of if-then-else conditional logic with the second one utilising nested if-then-else conditional logic.